

Impact of Internal and external Built Environment on Neighborhood Day-Night Land Surface Temperature: Insights from Unsupervised Fine-Grained Classification

Yuanfeng Ji ^{†a,b}, Chunxiao Wang ^{†c,d}, Jianhua Guo ^{a,b,e,*}, Jiashuo Wang ^f, Zhongchang Sun ^{a,b,e}, Dong Liang ^{a,b,e},
Huadong Guo ^{a,b,e}

^aInternational Research Center of Big Data for Sustainable Development Goals, , Beijing, 100094, , China

^bAerospace Information Research Institute, Chinese Academy of Sciences, , Beijing, 100094, , China

^cState Key Laboratory of Subtropical Building and Urban Science, Shenzhen University, , Shenzhen, 518060, , China

^dArchitecture and Urban Planning, Shenzhen University, , Shenzhen, 518060, , China

^eChina University of Chinese Academy of sciences, , Beijing, 101408, , China

^fSchool of Built Environment, University of New South Wales, , Sydney, 2052, , Australia

Abstract

Climate change exacerbates the urban heat island effect, thereby threatening urban neighborhoods. Current research on neighborhoods mainly focuses on the internal built environment, while the importance of the external built environment remains unclear. Incorporating external environmental factors into the analysis is crucial for identifying neighborhood morphological archetypes that are suitable for microclimate analysis and for gaining insight into the daytime and nighttime thermal environments of neighborhoods. This study obtained land surface temperature (LST) data from seven time periods, combined with multi-source geographic information, and used a PCA-based K-means clustering method to classify neighborhoods in the study area. Then, a random forest (RF) model and SHAP additive explanations were employed to analyze the contributions and nonlinear relationships of different factors. Finally, interaction analysis was used to compare the heterogeneous effects of different cooling strategies across various neighborhood types. The results show that: (1) Neighborhoods in the study area can be divided into five distinct categories, with significant differences in LST among the different categories. (2) The Straight-line Distance from the neighborhood to the city center (DIST) is the most important factor for both daytime and nighttime LST. (3) Different factors have varying effects across neighborhood types. This study proposes a refined fine-grained classification of neighborhoods and reveals the importance of external environmental factors, providing targeted suggestions for optimizing the thermal environment of neighborhoods, which is of great significance for building sustainable human settlements.

Keywords:

Urban heat island, Land surface temperature, Classification of Neighborhoods, K-means clustering, Heterogeneity effect

1. Introduction

Currently, global warming is leading to rising global temperatures, which in turn causes urban heat. Friesen and Taubenböck [1] found that the number of cities expected to exceed the mean annual temperature threshold of 29 °C will increase from 17 (2011–2040) to 217 (2071–2100), putting up to 320 million urban residents at risk. Urban warming thus threatens residents in urban neighborhoods. The urban heat island phenomenon exacerbates

^{††} These authors contributed equally to this work.

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Email addresses: jyfzz@bjfu.edu.cn (Yuanfeng Ji [†]), cxw@szu.edu.cn (Chunxiao Wang [†]), guojh@aircas.ac.cn (Jianhua Guo), jiashuo.wang@unsw.edu.au (Jiashuo Wang), sunzc@aircas.ac.cn (Zhongchang Sun), liangdong@radi.ac.cn (Dong Liang), hduo@radi.ac.cnn (Huadong Guo)

energy consumption in neighborhoods and threatens low-income neighborhoods [2]. High temperatures in the neighborhood also significantly increase the risk of illness among residents [3]. Therefore, it is necessary to conduct urban thermal environmental research at the neighborhood level to address the major challenges facing sustainable development.

A large number of studies have been conducted on the thermal environment at the neighborhood or block scale. Tang et al. [4] found that 3D and 2D landscape factors within blocks affect the land surface temperature (LST) of those blocks. Liu and Li [5] found that the socioeconomic status of a neighborhood affects the heat exposure of its residents. Zhao et al. [6] found that land use, human activities, and block functions affect the daytime and nighttime LST of blocks. However, most of these studies focus only on the effects of internal environmental factors, neglecting the influence of the external environment. The external environment of a neighborhood is not only important in planning [7] but is also considered an important factor correlated with the thermal environment. The ‘3-30-300 rules’ emphasize the importance of internal greening coverage and the accessibility of external greening [8]. In China, specific standards exist for planning the external environment of neighborhoods [9]. Current research also recognizes that urban landscapes exhibit thermal adjacency effects and affect the thermal environment of adjacent areas [10]. Arranging green spaces around neighborhoods can also reduce the heat risk for nearby residents [5]. Therefore, it is necessary to conduct further studies on the neighborhood thermal environment from a perspective that integrates both internal and external environments.

Classification is a common approach in urban thermal environment research. The Local Climate Zone (LCZ) [11] and Urban Functional Zone (UFZ) [12] are two classical classification methods. It helps us understand the heterogeneity of the urban microclimate and find targeted solutions. Zhou et al. [13] used LCZ to analyze the daytime and nighttime thermal environment of the core built-up area of Beijing. Zhao et al. [6] found significant differences in the daytime and nighttime thermal environments among different functional zones in Xi’an, China. The application of these two classical classification methods has advanced our understanding of the urban microclimate, but these methods still have shortcomings. For example, each LCZ should have a minimum diameter of 400–1,000 m [11], which limits the application of LCZ to smaller sites. UFZ essentially treats residential areas as a relatively homogeneous whole, making it difficult to reflect differences within the same UFZ. To address this gap, subsequent researchers have proposed other classification methods. Liu et al. [14] used a method combining LCZ and self-organizing maps (SOM) to classify neighborhoods. Joshi et al. [15] proposed a PCA-based K-means clustering approach to identify neighborhood morphological archetypes suitable for microclimate analysis. Sützl et al. [16] used K-means clustering to classify the thermal environment of neighborhoods in Zurich. However, these studies only consider the local environment of the neighborhood and fail to account for factors beyond the neighborhood boundary, such as the external environment [17, 18] and the location of the neighborhood [19]. As a small-scale landscape, the microclimate formation of urban neighborhoods may be more complex because it is regulated by multiple scales [20]. Therefore, we need to consider not only the local environment but also the external environment and more macro environmental factors to identify neighborhood prototypes suitable for thermal environmental analysis.

Moreover, many measures have been proposed to improve the urban thermal environment, including increasing tree canopy coverage [6] in urban areas, using high-albedo materials [21], and reasonably decreasing sky openness [22]. However, the priority of implementing these measures requires further study. The background built environment may affect the efficiency of cooling measures. Guo et al. [23] has shown that green infrastructure performs differently in various built environments. Kang et al. [17] has shown that landscape patterns play different roles in various LCZs. Therefore, we need to understand the performance of different factors in different types of neighborhoods. Understanding the heterogeneous effects of various measures can help us renovate our neighborhoods in a more targeted manner.

To fill these gaps, we take neighborhoods within the Fifth Ring Road in Beijing as a case study for further research. First, we constructed a framework incorporating indicators characterizing the internal and external environments of neighborhoods and used PCA-based K-means to cluster neighborhoods. Second, we use the random forest (RF) model and Shapley additive explanations (SHAP) to analyze the relative importance and nonlinear relationships of factors. Finally, through interaction analysis, we explore the heterogeneous effects of measures in various types of neighborhoods. Our research aims to achieve the following objectives: (1) incorporate internal and

external environmental characteristics of neighborhoods to achieve a fine-grained classification of neighborhoods within Beijing's Fifth Ring Road, and compare LST among different types of neighborhoods; (2) explore the relative importance of internal and external landscape factors on neighborhood LST, and the nonlinear relationships of some important factors; (3) systematically compare the roles of internal and external tree canopy, surface albedo, and sky openness in different types of neighborhoods. Through the above research, we hope to build a classification framework that integrates the internal and external environments of neighborhoods and conduct detailed research on the neighborhood thermal environment. We also hope to understand the importance of various landscape factors on the neighborhood thermal environment from a perspective that integrates internal and external environments, thereby providing targeted suggestions for thermal environment renovation for different types of neighborhoods in cities, supporting the sustainable development of urban neighborhoods.

2. Materials and Methods

2.1. Overall workflow of this study

The workflow of this study is shown in Fig. 1, which consists of five main steps. The first step is data preparation. The data used in this study include land surface temperature data, land use data, neighborhood boundary data, and other data. The data sources and processing methods are explained in detail in Section 2.3. The second step is indicator calculation and factor screening. From the above datasets, we calculated and obtained indicators covering multiple dimensions in order to comprehensively evaluate neighborhoods. We determined the optimal buffer zone range through the performance of machine learning models, selected variables characterizing the buffer zone built environment to build general linear models, and used the variance inflation factor (VIF) to screen out factors without severe multicollinearity. The third step is distinguishing the neighborhood prototype. We first performed dimensionality reduction on the factors using principal component analysis (PCA) with varimax rotation, then used the rotated components for K-means clustering analysis to classify neighborhoods. The fourth step is key factor analysis. We used the RF for modeling, optimized model parameters using Bayesian optimization, then used SHAP to analyze the relative importance of different factors, and used dependence plots to reveal the nonlinear relationships of key factors on the neighborhood thermal environment. The fifth step is heterogeneity analysis. We built models containing interaction terms between neighborhood group variables and environmental factors, observed the significance of the interaction terms, and combined forest plots and multiple comparisons to determine the effect of these factors across different neighborhood types. Finally, we propose targeted suggestions and measures for the thermal environmental optimization of neighborhoods.

2.2. Study area

Beijing is located in the northwestern part of the North China Plain ($39^{\circ}28' - 41^{\circ}25' \text{ N}$ and $115^{\circ}25' - 117^{\circ}30' \text{ E}$), with a total area of approximately 16,800 km². It is the capital of China, a political center, and an international exchange center, as well as one of the most populous cities in the world. According to the Köppen–Geiger climate classification, Beijing is in the humid continental climate (Dwa) climate zone [24]. The climate is characterized by hot, rainy summers, with the highest temperatures generally occurring in July. The monthly average temperature in plain areas is around 26°C, and the annual extreme maximum temperature is generally between 35°C and 40°C [25]. In summer, Beijing has long days and short nights, with sunrise occurring around 5:00 a.m. Summer twilight in Beijing is very long; sunset usually occurs after 19:30, but it does not get completely dark until around 21:30 [26]. The summer climate conditions and diurnal distribution in Beijing make it necessary to conduct more detailed research on its daytime and nighttime thermal environment.

The neighborhood in this study is shown in Fig. 2. This study focuses on the Fifth Ring Road area of Beijing, which is the urban core area and home to the majority of Beijing's residents. Its surface environment and three-dimensional building morphology are extremely complex, with different types of neighborhoods distributed [27], providing us with rich neighborhood samples for subsequent research. This area spatially overlaps with the main urban heat island effect area of Beijing, making it suitable for urban thermal environment research [28]. In summary, we conducted research on neighborhoods within the Fifth Ring Road in Beijing, covering a total of 2,725 neighborhoods.

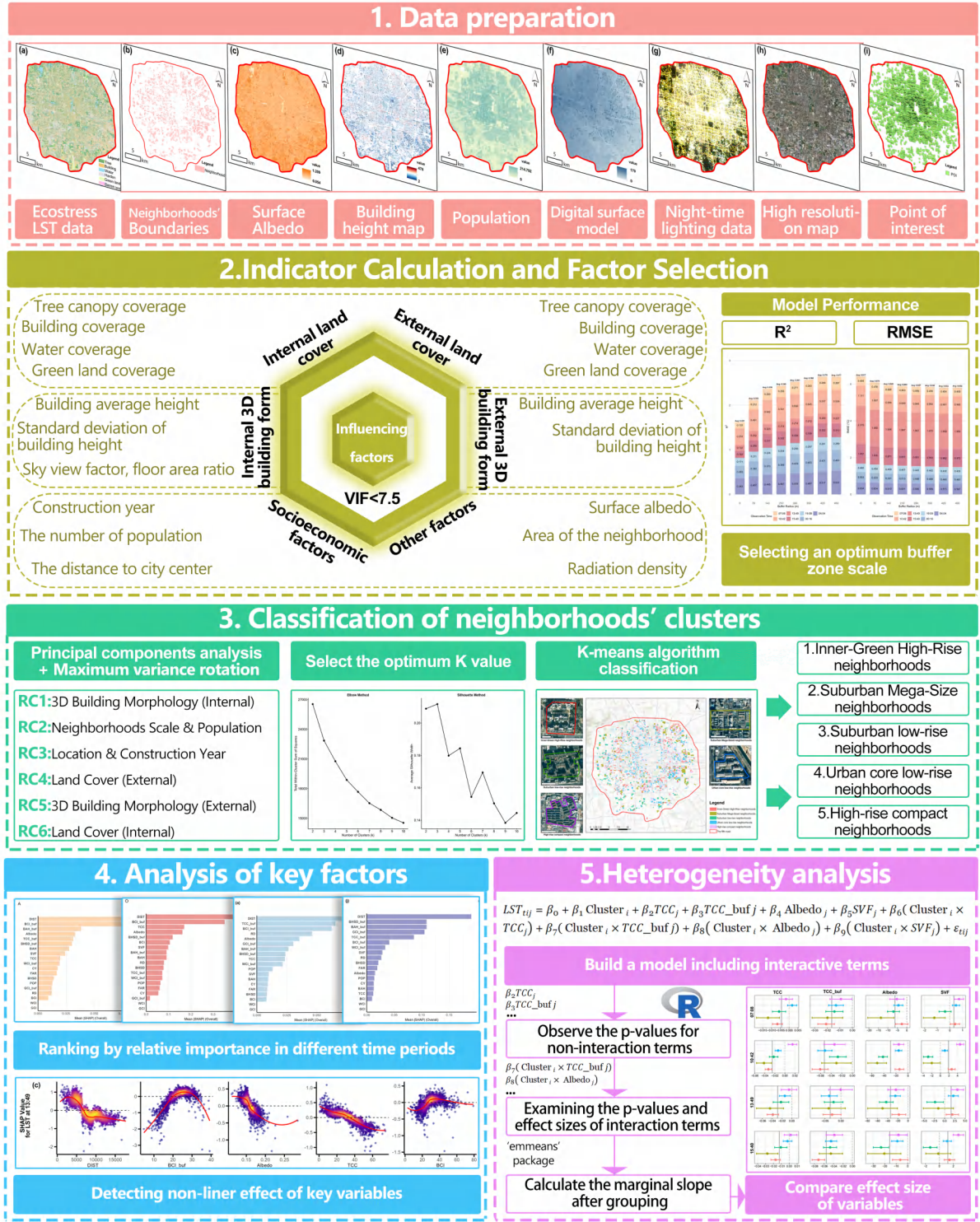


Figure 1: Research flowchart. The primary workflow of this study encompasses raw data acquisition, key variable selection, descriptive analysis, machine learning analysis, and geographical statistical analysis.

2.3. Data Sources

This study used multi-source geographic information data, including land surface temperature at different time periods, neighborhood boundary data, land cover maps, etc. These data are crucial for the detailed characterization of neighborhood attributes and further study of factors influencing the neighborhood thermal environment, laying the foundation for subsequent tasks such as constructing a reasonable framework of indicators, neighborhood classification, machine learning regression, and interaction analysis.

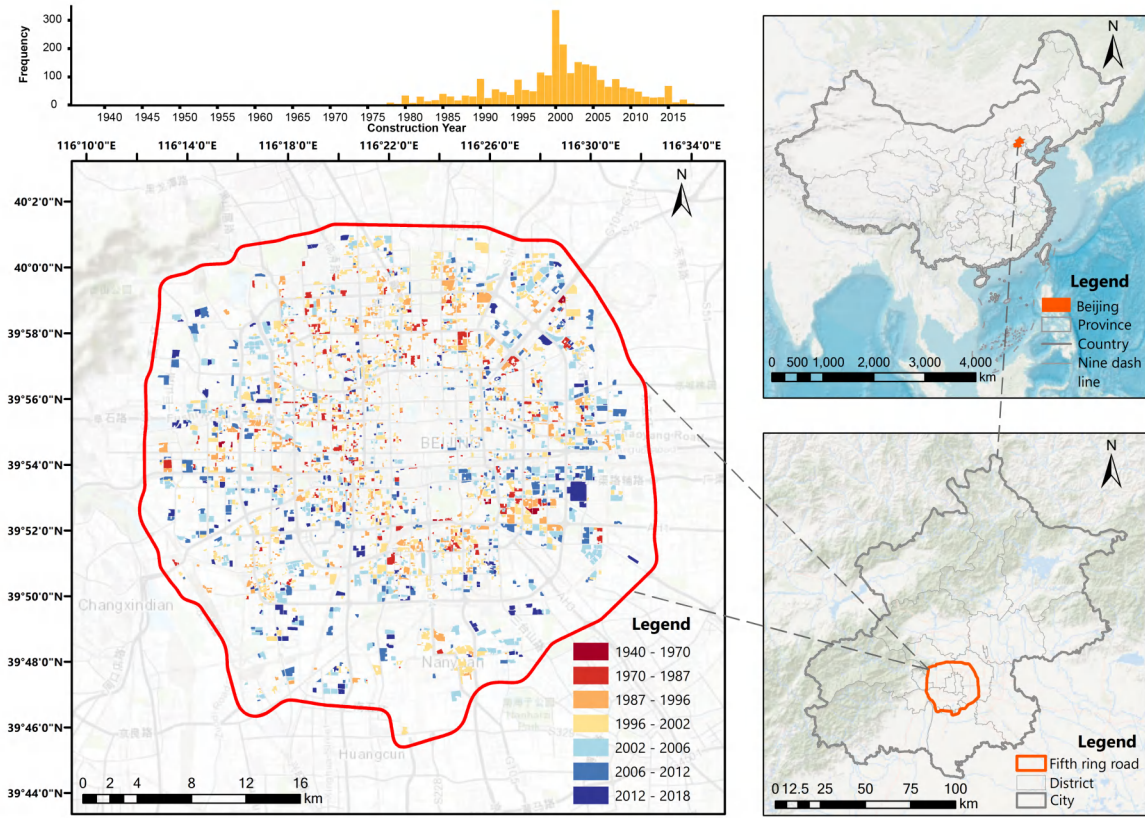


Figure 2: Beijing's fifth ring road area and selected neighborhoods in this study.

2.3.1. Land surface temperature

The land surface temperature (LST) data used in this study come from the Gridded Land Surface Temperature and Emissivity Instantaneous L2 Global 70m V002 of ECOSTRESS conducted by NASA. We obtained LST in the Beijing Fifth Ring Road area during summer at different daytime and nighttime periods from the ECOSTRESS dataset. For data selection, we followed these rules: (1) the acquisition dates of the images must be between June 1 and August 31; (2) The weather conditions on these dates were similar, including both completely clear skies and partly cloudy conditions.(3) The study area is not obscured by cloud cover.

These rules help us obtain daytime and nighttime LST data from different time periods, ensure consistency between LST data and other geographic data acquisition times, and avoid the interference of meteorological conditions. Moreover, selecting multiple images also helps enhance the stability of the results, making the analysis more reliable. We finally selected seven ECOSTRESS LST images that met the requirements. The shooting times were converted to Beijing time (UTC+8). The specific shooting times are shown in Table 1.

Table 1: Different time periods of ECOSTRESS images

Name of ECOSTRESS images	Date (YYYY/MM/DD)	Time (HH/mm)
ECOV002_L2T_LSTE_11103_009_50TMK_20200620T230803_0712_01	2020.6.21	07:08
ECOV002_L2T_LSTE_11896_008_50TMK_20200811T024201_0712_01	2020.8.11	10:42
ECOV002_L2T_LSTE_11774_009_50TMK_20200803T054914_0712_01	2020.8.3	13:49
ECOV002_L2T_LSTE_05370_009_50TMK_20190617T074029_0712_01	2019.6.17	15:40
ECOV002_L2T_LSTE_23095_016_50TMK_20220802T115933_0712_02	2022.8.2	19:59
ECOV002_L2T_LSTE_17026_008_50TMK_20210707T161600_0712_01	2021.7.8	00:16
ECOV002_L2T_LSTE_05580_006_50TMK_20190630T200402_0712_01	2019.7.1	04:04

2.3.2. Boundaries of neighborhoods

Our neighborhood boundaries come from the Baidu Map Open Platform(<https://lbsyun.baidu.com/>). First, we need to download point of interest (POI) data labeled as neighborhoods through the open API interface and clean the data. Second, we need to obtain POI information corresponding to the boundaries of areas of interest (AOI). Third, we need to vectorize and convert them to shapefile format, which contains the name and geographic location information of the neighborhoods. Finally, we select the AOI with the attribute of ‘residential area’(zhuzhai xiaoqu in Chinese) as the neighborhood boundary. Moreover, we also obtained the construction years of Beijing’s neighborhoods through the "Anjuke" website(<https://beijing.anjuke.com/>), a well-known real estate trading website in mainland China.

Then, we established a series of screening criteria to ensure that we obtained reliable neighborhood data. First, for consistency with the acquisition time of other data, we only selected neighborhoods built before 2019. Second, for data reliability, we removed neighborhood patches with too small an area; the neighborhoods included in this study have an area greater than 5,000 m². Third, to prevent the buffer zone from exceeding the boundary, we removed neighborhood data within a 500 m buffer of the Fifth Ring Road boundary. Finally, we selected 2,725 neighborhoods (Fig.2). Then, at 70 m intervals, we created buffers ranging from 70 m to 490 m around the neighborhoods to evaluate the environmental characteristics of the neighborhood externals.

2.3.3. Neighborhood’s characteristics

To comprehensively characterize the internal and external environments of neighborhoods, this study selected and calculated variables that represent multi-dimensional attributes—including land cover, 3D morphology, socioeconomic status, and physical properties—based on the LCZ framework [11], neighborhood classification studies [16, 14], and urban thermal environment research [27, 4]. Ultimately, 20 variables were selected to support our analysis. The definitions and formulas of these variables are presented in Table 2.

First, we derived eight land cover indicators (Table 2): four internal (tree canopy coverage TCC, building coverage BCI, water coverage WCI, greening coverage GCI) and four external (TCC_buf, BCI_buf, WCI_buf, GCI_buf) representing the corresponding proportions within the neighborhood and its buffer zone. To accurately reflect the fine-scale land cover of neighborhoods, we synthesized a high-resolution land cover dataset (Fig. ??a) from multiple open sources. We integrated Meta-WRI tree canopy maps [29], Baidu map open platform (<https://lbsyun.baidu.com/>), OpenStreetMap(<https://www.openstreetmap.org/>), and ESRI 10 m global land cover data [30]. In this dataset, greening includes grassland, farmland, and low vegetation. The integrated product achieved an overall accuracy of 89.2% based on 300 manually interpreted points (≥ 50 per class) from 0.6 m satellite imagery, acquired from Bigemap Pro in August 2020(<http://www.bigemap.com/>), confirming its suitability for neighborhood-scale analysis (Table S1).

Second, we calculated internal 3D building form indicators: average building height (BAH), standard deviation of building height (BHSD), and floor area ratio (FAR). External counterparts (BAH_buf, BHSD_buf) were computed within the buffer zone. We obtained the building height data in the study area through the Baidu Map Open Platform (<https://lbsyun.baidu.com/>). Through its application programming interface (API), we also retrieved the number of floors and height data for each building. Moreover, the sky view factor (SVF) was derived from 30 m ALOS DSM data (<http://www.eorc.jaxa.jp/ALOS/en/aw3d30/>). We utilized SAGA GIS to calculate SVF based on DSM data.

Third, we selected Construction year (CY), Population (POP), and Distance to city center (DIST) to describe the socioeconomic status of neighborhoods. CY was scraped from Anjuke (<https://beijing.anjuke.com/>). We parse and extract the Construction year using CSS selectors or XPath expressions. POP was obtained from WorldPop high-resolution (100 m) gridded data [31]. We acquired population data for the year 2020, which is consistent with the neighborhoods’ boundaries. DIST was computed as the straight-line distance from each neighborhood’s centroid to the Palace Museum (city center) using ArcGIS Pro’s proximity analysis.

Fourth, Surface albedo(Albedo), area of neighborhoods(Area) and mean radiance intensity (MR) were selected to represent the physical properties of neighborhoods. Albedo was derived from a cloud-free Landsat 8/9 image that was taken on 3 August 2020. Using the method of Liang [32], the 30 m shortwave albedo over Beijing’s Fifth Ring Road area was calculated as:

$$Albedo = 0.356\rho_2 + 0.130\rho_4 + 0.373\rho_5 + 0.085\rho_6 + 0.072\rho_7 - 0.0018, \quad (1)$$

where ρ_2 is the blue band (Landsat 8 Band 2), ρ_4 is the red band (Band 4), ρ_5 is the near-infrared (NIR, Band 5), ρ_6 is the shortwave infrared 1 (SWIR1, Band 6), and ρ_7 is the shortwave infrared 2 (SWIR2, Band 7).

Based on the boundaries of each neighborhood, we used the Geometry Calculator tool in ArcGIS Pro to calculate the area of each neighborhood. Then, Nighttime light data were sourced from the Sustainable Development Science Satellite 1 (SDGSAT-1). It can provide 10m resolution nighttime light data, reaching about 1m resolution in some areas. Using the `terra` package in R, we extracted the HDR band and calculated the average radiance per neighborhood. The average radiation brightness of a neighborhood is calculated as:

$$\bar{R} = \frac{\sum_{i=1}^n DN_i}{n} \quad (2)$$

where $\bar{R}$ represents the average radiation brightness of the neighborhood; DN_i is the brightness value (HDR band value) of the i -th pixel within the neighborhood; and n is the total number of valid pixels covered by the neighborhood.

2.4. Methods

This study systematically integrates unsupervised classification, machine learning, and interaction analysis to propose an integrated analytical framework for classifying neighborhoods in Beijing's core urban area and exploring the impacts of different factors on daytime and nighttime LST. The methodology consists of three main steps. First, dimensionality reduction is performed using PCA, followed by neighborhood classification via the K-means algorithm. Second, to interpret internal and external environmental variables, we construct a random forest model and use SHAP analysis to assess the relative importance of factors and the nonlinear effects of key factors. Third, for heterogeneity analysis, we calculate the marginal slopes of the same factor across different groups and apply multiple comparisons to examine the heterogeneous effects of various factors. This approach provides a detailed strategy for understanding differences among neighborhoods, the contributions of internal and external environments to the thermal environment, and the heterogeneity of these effects.

2.4.1. Neighborhoods classification based on PCA and K-means method

we chose to use principal component analysis (PCA) to process these variables. PCA can reduce variable dimensions and eliminate multicollinearity while retaining the original data information as much as possible, facilitating subsequent clustering and interpretation. As shown in Fig. S2, the correlation between some variables exceeds 0.8, which confirms the validity of our use of PCA. After that, we need to select appropriate principal components and explain their meanings. According to the Kaiser criterion [33], we selected the first six principal components with eigenvalues >1 for subsequent analysis. They together explain 61.1% of the variance. Fig. S3 shows the variance explained and the eigenvalues of the first ten principal components. Factor loadings represent the contribution and correlation of each original variable to certain principal components (PCs). Varimax rotation is an orthogonal rotation method that makes the practical meaning of principal components clearer and easier to name, which converts PCs to rotated components (RCs). We combined the practical meaning of the variables to provide reasonable interpretations for different principal components (Table 3). Table S3 shows the results of principal component loadings and varimax rotation loadings.

To reasonably classify neighborhoods with different characteristics, we used the K-means algorithm to cluster the six RCs that characterize neighborhood features. The algorithm divides grid cells into k clusters, with each cell belonging to the cluster with the nearest centroid [16]. This algorithm has been successfully applied in studies on urban streets and neighborhood classification related to the thermal environment [34, 16].

Choosing an appropriate k value is crucial for the rationality of the classification. This study used the elbow method and silhouette coefficient to help determine the optimal k value. The elbow method is a graphical method [35]. We plotted the within-cluster sum of squares (WCSS) curve for different k values. The silhouette coefficient [36] is used to measure the similarity of an object to its own cluster compared to other clusters. The closer the silhouette coefficient is to 1, the better the classification effect. The silhouette coefficient is calculated as follows:

Table 2: Indicators and calculations

Category	Indicator	Meaning	Formula / Source
Internal land cover	TCC	Measurement of tree canopy coverage in the neighborhood.	$TCC = \frac{S_{tree}}{S_{total}}$
	BCI	Measurement of building coverage in the neighborhood.	$BCI = \frac{S_{building}}{S_{total}}$
	WCI	Measurement of water coverage in the neighborhood.	$WCI = \frac{S_{water}}{S_{total}}$
	GCI	Measurement of greening coverage in the neighborhood.	$GCI = \frac{S_{green}}{S_{total}}$
External land cover	TCC_buf	Measurement of tree canopy coverage in the buffer zone.	$TCC_buf = \frac{S_{tree_buf}}{S_{total_buf}}$
	BCI_buf	Measurement of building coverage in the buffer zone.	$BCI_buf = \frac{S_{building_buf}}{S_{total_buf}}$
	WCI_buf	Measurement of water coverage in the buffer zone.	$WCI_buf = \frac{S_{water_buf}}{S_{total_buf}}$
	GCI_buf	Measurement of greening coverage in the buffer zone.	$GCI_buf = \frac{S_{green_buf}}{S_{total_buf}}$
Internal 3D building form	BAH	Measurement of building average height in the neighborhood.	$BAH = \frac{\sum_{i=1}^n H_i}{n}$
	BHSD	Standard deviation of building height in the neighborhood.	$BHSD = \sqrt{\frac{\sum_{i=1}^n (H_i - \bar{H})^2}{n}}$
	FAR	Measuring development intensity in the neighborhood.	$FAR = \frac{\sum_{i=1}^n (S_i \times F_i)}{S_{total}}$
	SVF	Mean value of sky view factor in the neighborhood.	Calculated in SAGA GIS
External 3D building form	BAH_buf	Building average height in the buffer zone.	$BAH_buf = \frac{\sum_{i=1}^{n_{buf}} H_{i,buf}}{n_{buf}}$
	BHSD_buf	Standard deviation of building height in buffer zone.	$BHSD_buf = \sqrt{\frac{\sum_{i=1}^{n_{buf}} (H_{i,buf} - \bar{H}_{buf})^2}{n_{buf}}}$
Socioeconomic factors	CY	Construction year of the neighborhoods.	Anjuke website
	POP	Population number in the neighborhood.	Worldpop
	DIST	Straight distance to the city centre.	$DIST = \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2}$
Physical properties	Albedo	Average surface shortwave albedo in neighborhood.	Formula (1)
	Area	Area of the neighborhood.	S_{total}
	RD	Radiation density per unit area in neighborhood.	Formula (2)

Note: S_{total} : total area of neighborhoods; S_{tree} : tree area in neighborhoods; $S_{building}$: building area in neighborhoods; S_{water} : water area in neighborhoods; S_{green} : greening area(exclude tree coverage) in neighborhoods; S_{total_buf} : area of buffer zone; S_{tree_buf} , $S_{building_buf}$, S_{water_buf} , S_{green_buf} : tree, building, water, greening area in buffer zone; H_i : height of building i ; n : number of buildings; $\bar{H}$: average building height; S_i : floor area of building i ; F_i : floor number of building i ; n_{buf} : building count in buffer zone; $H_{i,buf}$: height of building i in buffer zone; $\bar{H}_{buf}$: average building height in buffer zone; (x_i, y_i) : geometric center coordinates of neighborhood; (x_0, y_0) : geometric center coordinates of Beijing city center(Palace Museum).

$$\text{Silhouette Score} = \frac{b - a}{\max(a, b)} \quad (3)$$

where a is the average distance between a data point and all other points in the same cluster (inter-cluster distance) and b is the average distance between a data point and all other points in the nearest cluster (inter-cluster distance).

As shown in Fig.3, WCSS starts at a high point at $k=2$ and then declines rapidly. The decline rate begins to slow down gradually at $k \geq 5$. The silhouette score peaks at $k=3$ and reaches a secondary peak at $k=5$. In summary, $k=5$ is the turning point where the decline in WCSS begins to slow down. Although it is not the highest point of the silhouette score, it remains at a relatively high level. Therefore, $k=5$ is the best choice that balances classification

Table 3: Practical meaning of each principal component

Rotated Component	Core Representation	Dominant Indicators	Interpretation
RC1	3D Building Morphology (Internal)	BAH (−0.515), BHSD (−0.455), FAR (−0.468), SVF (0.438)	Reflects building density and spatial openness within the neighborhood.
RC2	Neighborhoods Scale & Population	POP (−0.697), Area (−0.678)	Represents the neighborhood's land area and population agglomeration level.
RC3	Location & Construction Year	Albedo (−0.602), DIST (0.524), CY (−0.452)	Reflects the neighborhood's urban location, surface property, and construction age.
RC4	Land Cover (External)	WCI_buf (−0.643), TCC_buf (−0.553), BCI_buf (−0.400)	Characterizes the composition of water, vegetation, and built-up area in the external buffer zone.
RC5	3D Building Morphology (External)	BAH_buf (−0.642), BHSD_buf (−0.639)	Describes building height and height heterogeneity in the external buffer.
RC6	Land Cover (Internal)	TCC (−0.627), BCI (0.595)	Shows the spatial pattern of vegetation and built-up area inside the neighborhood.

effectiveness and fine morphological division. Thus, this study selects $k=5$ as the optimal number of clusters for subsequent research.

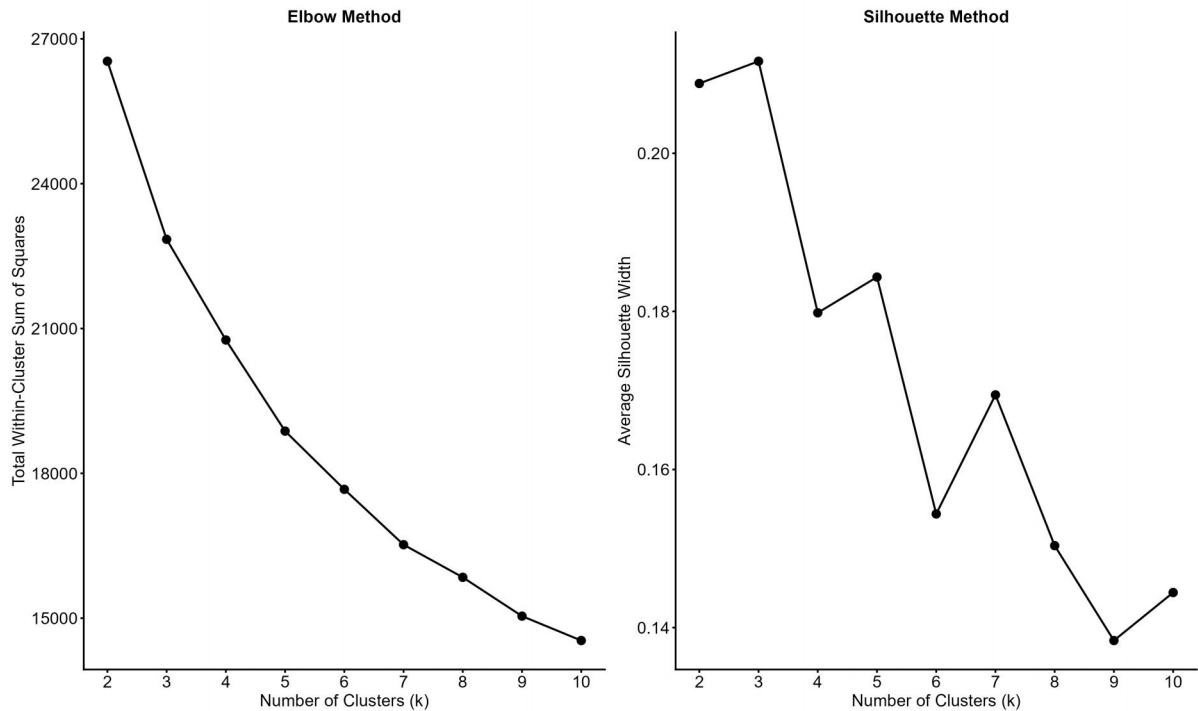


Figure 3: Determining the optimal K value with Elbow method and Silhouette method.

After determining the optimal k value, we used the K-means algorithm to classify neighborhoods reasonably. We then calculated the average scores of different types of neighborhoods on the six rotated RCs and used this to judge the characteristics and practical significance of each cluster through the average score of RCs. During this process, a small number of samples in the classified dataset were automatically disregarded by the algorithm (8 samples).

Subsequently, we used the Kruskal–Wallis test to check whether the average LST of each cluster is significantly

different overall, and then performed pairwise comparisons using the Wilcoxon rank-sum test, correcting the p-values of multiple comparisons using the Benjamini–Hochberg method, which can reduce the probability of false positives in multiple comparisons [37]. We used the compact letter display (CLD) algorithm [38] to conduct the multiple comparisons of LST among different neighborhood types, which can help us find heterogeneity in the thermal environment among different types of neighborhoods.

2.4.2. Explaining internal and external environmental factors based on RF and SHAP

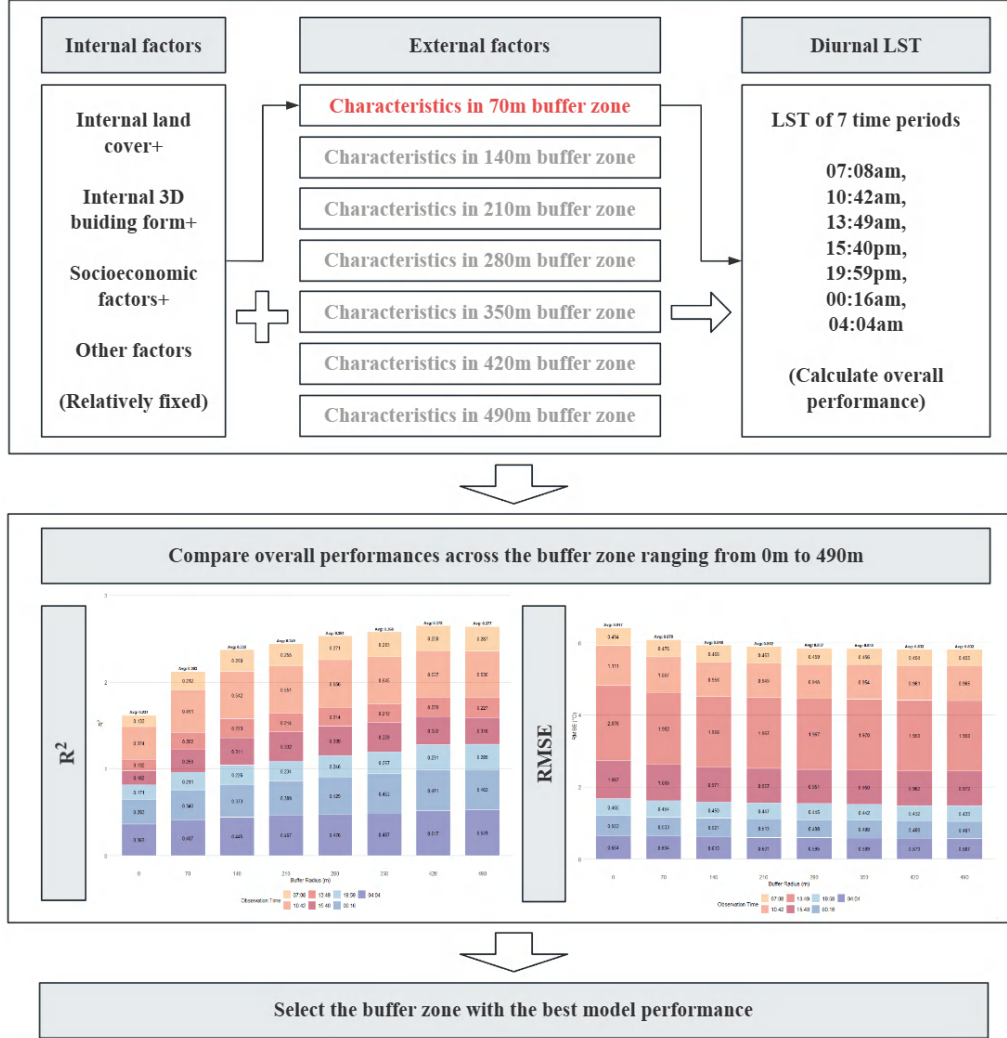


Figure 4: Determine the optimal buffer zone distance by comparing the models' overall performance in terms of daytime and nighttime LST. External factors were incorporated into the model separately to evaluate the overall model performance, which represent the environmental characteristics within 70–490 m buffer zones external the neighbourhood. Only one set of buffer zone variables was included in each round of overall performance evaluation. Notably, the baseline model with a 0-metre buffer contained no external factors.

The RF algorithm is an ensemble learning method based on decision trees, improving model accuracy and stability by aggregating multiple decision trees [39]. First, we use RF models to select the optimum buffer distances. This process is shown in Fig. 4. This study used an RF model with fixed parameters (ntree = 500, min.node.size = 5) to output model performance (R^2 , RMSE) for different buffers (0m–490m, 70m intervals). Among them, the internal factors were fixed. We sequentially incorporated buffer zone environmental characteristics at different distances and compared the diurnal performance of models that included different buffer variables to determine the optimal buffer distance for this study. Finally, we determined 420 m as the optimal buffer range for this study. The performance of the models is shown in Fig. S1.

Then, we constructed a full model incorporating both internal and buffer zone built environment features for subsequent analysis. Hyperparameter optimization was an important task. We used Bayesian optimization with

RMSE as the evaluation criterion for model performance and set parameter ranges. The search range for ntree was 100–2000, for mtry was 1–19, and for min.node.size was 1–9. Bayesian networks can automatically search for the best parameter combination to achieve hyperparameter optimization for different models. After that, we used the SHAP method to explain the optimized models.

In this study, we use Shapley Additive Explanations (SHAP) to explain the contributions and nonlinear relationshipS of factors. SHAP is a method that can effectively explain the contributions of feature variables on the dependent variable. This explanation method extends the concept of Shapley values from game theory [40], which aims to fairly distribute the contributions of players when they jointly achieve a certain outcome. Shapley values can be used in machine learning to quantify the contribution of each feature. The Shapley value in the model is given by:

$$\text{Shapley}(X_j) = \sum_{S \subseteq N \setminus \{j\}} \frac{k!(p-k-1)!}{p!} (f(S \cup \{j\}) - f(S)), \quad (4)$$

where p is the total number of features, $N \setminus \{j\}$ is a set of all possible combinations of features excluding X_j , S is a feature set in $N \setminus \{j\}$, $f(S)$ is the model prediction with features in S , and $f(S \cup \{j\})$ is the model prediction with features in S plus feature X_j .

2.4.3. Comparing heterogeneity of factor effect sizes using interaction analysis

In this study, We conduct an interaction analysis to examine whether the associations between environmental variables and LST varied across neighborhood types. Specifically, interaction terms between neighborhood type and environmental predictors are incorporated into linear regression models, and estimated marginal slopes are derived to quantify heterogeneous environmental effects across neighborhoods. Pairwise comparisons of marginal slopes are subsequently performed using Tukey-adjusted contrasts, and compact letter display (CLD) grouping is applied to facilitate the interpretation of inter-group differences.

We incorporate these variables into models and constructed models containing interaction terms. The model constructed in this study is shown in the following formula:

$$LST_{tij} = \beta_0 + \beta_1 \text{Cluster}_i + \beta_2 X_j + \beta_3 (\text{Cluster}_i \times X_j) + \varepsilon_{tij} \quad (5)$$

where LST_{tij} represents the average land surface temperature (LST) of the neighborhood at different times; Cluster_i is a categorical variable representing neighborhood type, modeled using dummy coding with one type treated as the reference category; X_j represents one of the four environmental variables (TCC, TCC_buf, Albedo, SVF); β_0 represents the intercept term. β_1 – β_3 are the regression coefficients; and ε_{tij} is the random error term. Separate models were fitted for each predictor variable and each time period.

For a given neighborhood type c , the marginal slope of X on Y across different groups is defined as the partial derivative of the conditional expectation:

$$\text{Slope}_c(X) = \frac{\partial \mathbb{E}[Y | X, \text{Type} = c]}{\partial X} = \begin{cases} \beta_X, & c = \text{reference}, \\ \beta_X + \delta_i, & c = \text{non-reference } i. \end{cases} \quad (6)$$

where Y denotes the land surface temperature (LST) at a given time period; X represents one of the continuous predictors of interest (TCC, TCC_buf, Albedo, SVF); $\text{Type} = c$ indicates that the neighborhood belongs to cluster c (with $c = 1, \dots, 5$ covering the five identified neighborhood types); β_X is the main effect coefficient of X from the linear model (Equation 5); and δ_i is the interaction coefficient between X and the i -th non-reference type (with $i = 1, \dots, 4$ indexing the four non-reference types).

These marginal slopes were estimated using the `emmeans` function from the `emmeans` package in R 4.5.3 [41]. This approach estimates the conditional effect of each environmental variable on LST within different neighborhood types while accounting for the fitted interaction structure of the regression model. Standard errors and 95% confidence intervals were derived from the variance–covariance matrix of the estimated regression coefficients. To assess whether the environmental effects differed significantly among neighborhood types, pairwise comparisons

of the estimated marginal slopes were performed with Tukey adjustment for multiple testing, thereby controlling the family-wise error rate. Finally, compact letter display (CLD) grouping [38] was applied to summarize the pairwise comparison results, where neighborhood types sharing the same letter were considered not significantly different, while groups with different letters indicated statistically significant differences. The CLD results were further visualized in the forest plots to facilitate the interpretation of effect heterogeneity across neighborhood types.

3. Results

3.1. Descriptive analysis

Table 4: Statistical scope of variables

Variable	Min-Max	Mean	Median	Standard Deviation
Land surface temperature (°C)				
LST 00:16	20.770 – 25.550	23.639	23.700	0.639
LST 04:04	19.680 – 25.110	23.000	23.144	0.812
LST 07:08	25.603 – 29.930	27.564	27.557	0.537
LST 10:42	34.249 – 43.357	39.397	39.434	1.401
LST 13:49	27.511 – 52.786	45.998	46.259	2.306
LST 15:40	30.117 – 39.350	34.882	34.907	1.150
LST 19:59	22.279 – 31.220	29.814	29.862	0.581
Neighborhood characteristics				
TCC (%)	0.700 – 74.350	31.873	31.900	13.277
BCI (%)	2.840 – 79.100	29.297	28.590	8.567
WCI (%)	0.000 – 6.700	0.011	0.000	0.196
GCI (%)	0.000 – 0.570	0.000	0.000	0.011
CY	1940 – 2018	1999.839	2001.000	8.972
BAH (m)	6.000 – 231.571	54.196	45.679	30.998
BHSD (m)	0.000 – 166.300	30.888	27.692	20.971
SVF	0.329 – 0.980	0.716	0.730	0.117
Albedo	0.112 – 0.285	0.159	0.158	0.014
POP	62.185 – 32878.639	1759.455	1074.000	1985.554
DIST	729.309 – 17744.654	7925.000	7699.000	3100.000
FAR	0.092 – 33.064	5.671	4.727	3.311
MR(W m ⁻² sr ⁻¹)	1.853 – 1230.263	25.400	15.100	54.200
TCC_buf (%)	7.310 – 54.480	30.308	30.170	5.989
BCI_buf (%)	4.710 – 39.270	22.749	22.980	4.843
WCI_buf (%)	0.000 – 24.060	0.873	0.000	1.987
GCI_buf (%)	0.000 – 27.230	0.337	0.000	1.611
BAH_buf (m)	8.779 – 90.658	37.099	37.100	12.490
BHSD_buf (m)	3.940 – 99.193	35.658	35.827	12.376
Area (m ²)	5012.685 – 1049583.525	36628.719	20621.581	46599.463

Table 4 presents the statistical ranges of the variables. For land surface temperature (LST), the mean values differ markedly across time periods: the lowest occur at midnight (00:16, 23.639°C) and pre-dawn (04:04, 23.000°C), while the afternoon (13:49) records the highest values.

Regarding neighborhood characteristics, the internal blue-green space is dominated by uneven tree canopy coverage. Although the mean tree canopy coverage (TCC) is 31.873%, its standard deviation (13.277%) indicates large disparities among neighborhoods. In contrast, water bodies (WCI, mean 0.011%) and greening (GCI, near 0%) are extremely scarce. Building height also varies considerably: the mean building average height (BAH) is 54.196 m (SD 30.998 m), and the standard deviation of building height (BHSD) averages 30.888 m, reflecting a mixture of low-rise and super-high-rise buildings. Development intensity, measured by floor area ratio (FAR), shows substantial variation (SD = 3.311). Population (POP) is similarly uneven, with a mean of 1,759.455 and a standard

deviation of 1,985.554. In terms of physical properties, surface albedo is relatively homogeneous across neighborhoods (mean = 0.159, SD = 0.014). Nighttime mean radiance intensity (MR) averages $25.400\text{W m}^{-2}\text{ sr}^{-1}$ (SD= $54.200\text{W m}^{-2}\text{ sr}^{-1}$). Turning to the external environment, the mean proportions of building coverage (BCI_buf) and tree canopy (TCC_buf) around neighborhoods are 22.749% and 30.308%, respectively, while greening (GCI_buf) and water bodies (WCI_buf) account for only small fractions. External building height (BAH_buf) and its standard deviation (BHSD_buf) have standard deviations of 12.490 m and 12.376 m, respectively, suggesting notable heterogeneity in the three-dimensional form of surrounding areas.

In summary, LST in the study area exhibits a pronounced diurnal variation, and there are significant spatial differences in both internal and external built environments.

Table 5: Average scores for rotated components across different clusters of neighborhoods

Categories	RC1	RC2	RC3	RC4	RC5	RC6	Proportion
Inner-Green High-Rise	-0.727	-0.035	0.016	-0.369	-0.937	-0.918	26.1%
Suburban Mega-Size	0.098	-4.200	0.783	-0.114	-0.062	-0.779	5.5%
Suburban low-rise	1.210	-0.206	2.000	0.140	1.250	-0.439	15.1%
Urban core low-rise	1.210	0.474	-0.872	0.210	0.429	0.424	36.5%
High-rise compact	-2.620	0.579	-0.184	0.027	-0.579	1.160	16.8%

Note: RC1: 3D Building Morphology (Internal); RC2: Neighborhoods Scale & Population; RC3: Location & Construction Year; RC4: Land Cover (External); RC5: 3D Building Morphology (External); RC6: Land Cover (Internal).

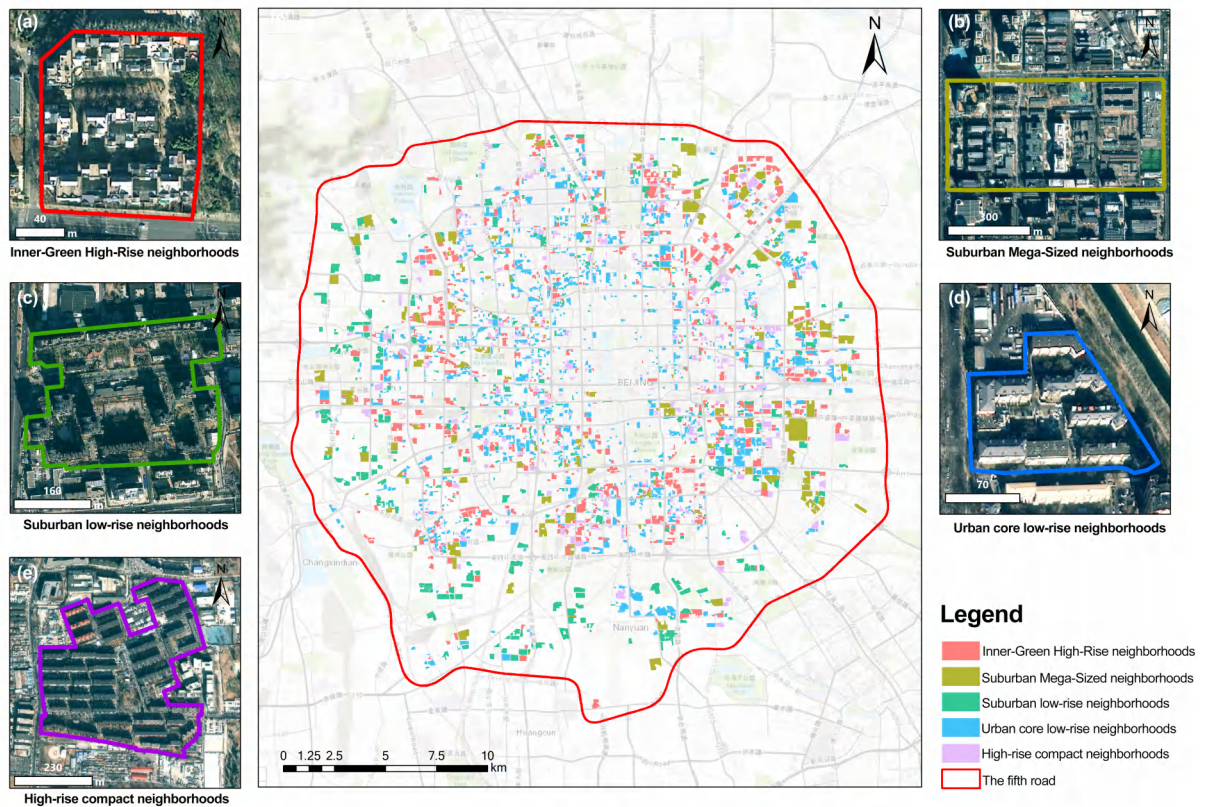


Figure 5: Different types of neighborhoods.(a) Inner-Green High-Rise neighborhoods. (b)Suburban Mega-Sized neighborhoods.(c)Suburban low-rise neighborhoods.(d)Urban core low-rise neighborhoods.(e)High-rise compact neighborhoods.

3.2. The classification of neighborhoods

3.2.1. Different types of neighborhoods and their characteristics

Table 5 shows the mean scores of each principal component in each type of neighborhood after classification. We interpret the characteristics of each category based on the practical meaning of each principal component. The neighborhoods can be divided into five categories: Inner-Green High-Rise neighborhoods, Suburban Mega-Sized neighborhoods, Suburban low-rise neighborhoods, Urban core low-rise neighborhoods, and High-rise compact neighborhoods. Fig.5 shows samples of location and satellite images of different types of neighborhoods.

Inner-Green High-Rise neighborhoods account for 26.1% and are mainly distributed between the Third and Fifth Ring Roads. They feature relatively high average building heights and high tree coverage. Suburban Mega-Sized neighborhoods represent a smaller proportion (5.5%), primarily located in the eastern area between the Fourth and Fifth Ring Roads. They are characterized by large areas, dense populations, and distance from the city center. Suburban low-rise neighborhoods account for 15.1% and are mainly distributed in the southwestern area between the Fourth and Fifth Ring Roads. They are far from the city center and have low-rise buildings both internally and externally. Urban core low-rise neighborhoods are one of the dominant types in this study, accounting for 36.5%. They are mainly distributed between the Second and Fifth Ring Roads and are characterized by older construction years, small areas, low average building heights, and low tree canopy coverage. High-rise compact neighborhoods account for 16.8% and are mainly distributed in the eastern area between the Second and Fourth Ring Roads. They feature high internal building heights, small areas, and low tree canopy coverage.

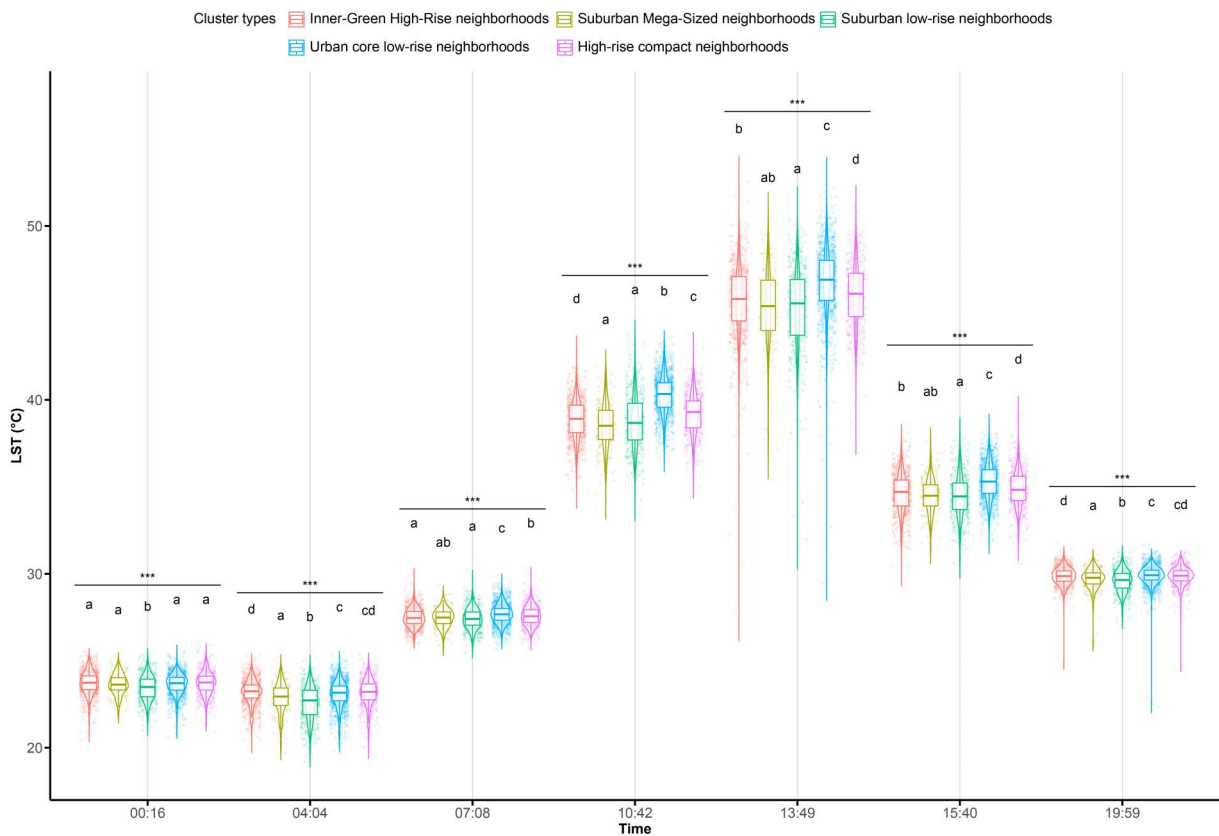


Figure 6: Comparison of LST of different types of neighborhoods over seven time periods. Groups not sharing the same letter are considered to have significant differences.

3.2.2. Differences in thermal environment among various types of neighborhoods

Subsequently, the Benjamini–Hochberg method was used for more detailed pairwise comparisons of LST among different neighborhoods, with results shown in Fig.6. The study found that the five types of neighborhoods show significant differences in LST across all time periods. During the morning and afternoon periods (10:42–15:40), LST differs more markedly among different neighborhood types. Urban core low-rise neighborhoods have significantly higher LST during this period than other types of neighborhoods, followed by High-rise

compact neighborhoods. Inner-Green High-Rise neighborhoods have slightly higher LST during this period than Suburban Mega-Sized neighborhoods and Suburban low-rise neighborhoods, which have lower LST during this period and do not show significant differences between them. During nighttime and early morning (00:16–07:08, 19:59), although the LST of each neighborhood type shows statistically significant differences, the differences are generally small.

3.3. Relative importance and nonlinear relationship of factors in seven time periods

3.3.1. Relative importance during daytime periods

Fig.7 shows the relative importance and summary plots of 19 variables ('Area' is not included in the regression) for LST during daytime periods. We found that both internal and external environmental factors contribute to the LST. DIST and BCI_buf are the main contributors to LST, and in most diurnal periods, their mean absolute SHAP values far exceed those of other factors. DIST has a negative correlation with LST, while BCI_buf has a positive correlation. Albedo, TCC, BHSD_buf, and TCC_buf can be considered second-tier contributors. These factors have negative correlations with LST during the daytime. In contrast, the contributions of WCI and GCI are weak, likely due to the fact that water bodies and large areas of greening are rare in Beijing's neighborhoods.

In summary, during the daytime, the DIST and BCI_buf are the most important factors affecting LST; both external and internal factors are important contributors to the LST in the neighborhood. This result explains the importance of the location, internal environment, and external environment of neighborhoods, providing new insights for subsequent research on the summer daytime thermal environments of neighborhoods.

3.3.2. Relative importance during nighttime periods

Fig.8 shows the relative importance of 19 variables for LST during nighttime periods. The study found that DIST, BCI_buf, BAH_buf, and BHSD_buf can be considered first-tier contributors to nighttime LST. DIST remains one of the most important factors during nighttime periods. Compared with daytime periods, the importance of the external environment of neighborhoods is much greater than that of the internal environment during nighttime. In the evening (19:59), BCI_buf is still the most important factor. During midnight (00:16) and dawn (04:04), the average height and standard deviation of building height (BAH_buf and BHSD_buf) become the most important factors for LST. The external land cover and internal three-dimensional building form can be considered second-tier contributors. For example, GCI_buf and WCI_buf rank fourth in importance at 00:16 and 04:04, respectively. Internal three-dimensional form factors, such as FAR and SVF, are also important factors. The direction of SVF is opposite to that during the daytime, with SVF having a cooling correlation with nighttime LST, reflecting the complex role of building three-dimensional morphology on daytime and nighttime LST.

In summary, at night, among the top 10 variables in importance ranking, variables characterizing the external environment occupy most of the positions. DIST is still the most important factor. The importance of 3D external building morphology becomes significant at night. This finding helps us understand the role of location and the external built environment on the summer nighttime thermal environment of neighborhoods, enabling further research.

3.3.3. Nonlinear relationships of key internal and external factors across time periods

The study used SHAP dependence plots to investigate the nonlinear relationships of the top five key factors for each time period on LST. The results are shown in Fig.9. The study found nonlinear relationships between key factors and the LST in neighborhoods.

In this study, we found that the DIST, as a key variable affecting both daytime and nighttime LST, shows a decreasing trend during daytime periods. For example, at 10:42, the SHAP value decreases continuously as DIST increases; when $DIST \geq 7500m$, the SHAP value turns from positive to negative, indicating that DIST then has a negative contribution to LST. At night, DIST and SHAP values show more complex nonlinear relationships. At 00:16, the SHAP value first decreases with increasing DIST, then when $DIST \geq 5000m$, the SHAP value increases with increasing DIST, and when $DIST \geq 10000m$, the SHAP value decreases again with increasing DIST. The construction year of the neighborhood also shows a threshold effect. For example, at 10:42, when $CY \geq 2000$, the SHAP value drops rapidly.

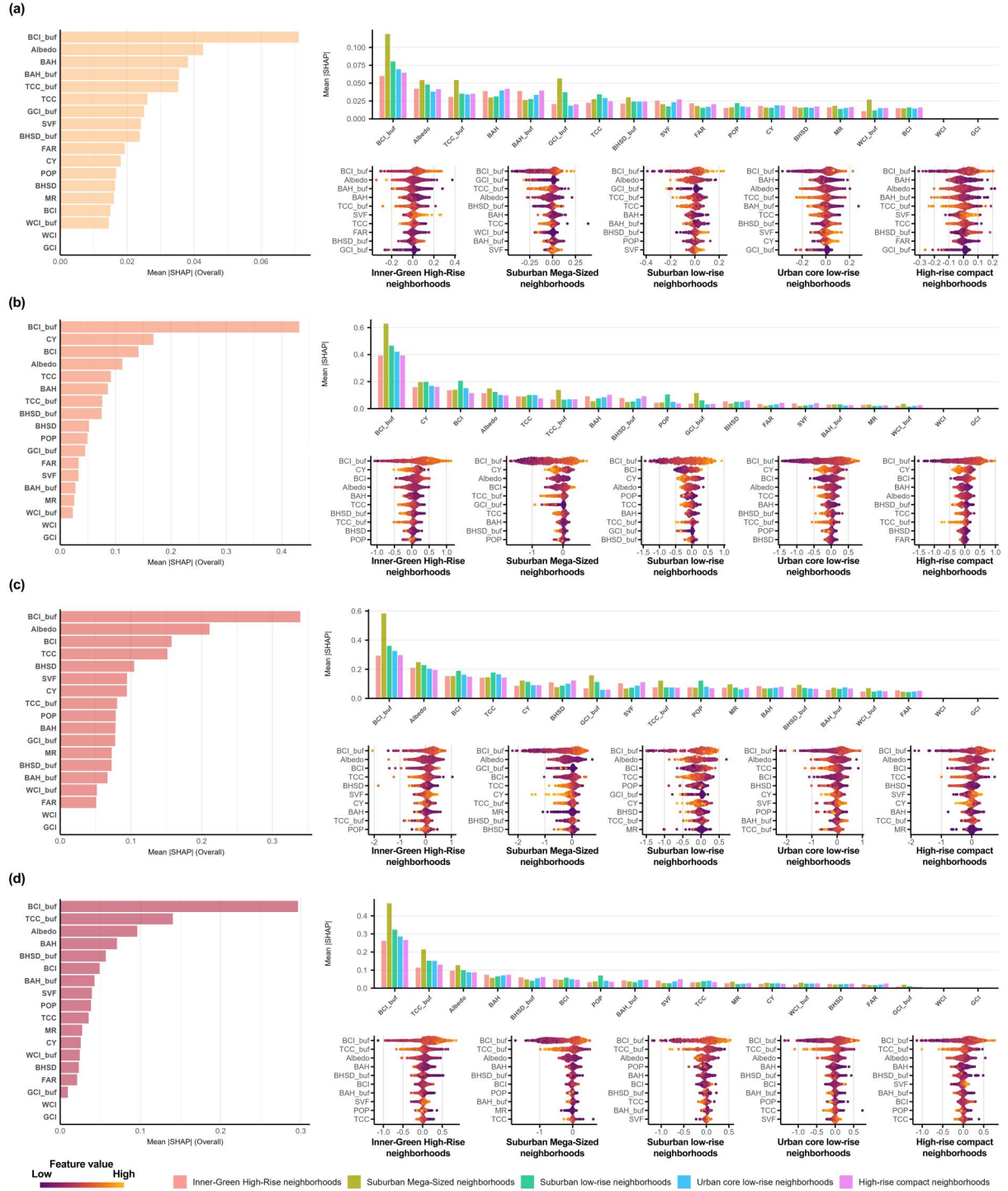


Figure 7: Relative importance and summary plots of different variables during daytime periods. (a) 07:08; (b) 10:42; (c) 13:49; (d) 15:40.

The external land cover surrounding neighborhoods profoundly affects the internal daytime and nighttime LST of neighborhoods. BCI_buf is a clear warming factor. For example, at 15:40, when BCI_buf ≥ 20 , the SHAP value changes from negative to positive. External tree canopy coverage (TCC_buf) has a cooling correlation with neighborhood LST. During both daytime and nighttime (15:40, 04:04), it shows a significant negative correlation with the SHAP value; when TCC_buf ≥ 35 , the SHAP value turns from positive to negative.

The role of the three-dimensional building morphology around neighborhoods on the daytime and nighttime thermal environments of neighborhoods is also complex. BAH_buf shows opposite effects during the daytime and nighttime: during the day, BAH_buf exhibits a cooling effect. For example, at 07:08, an increase in BAH_buf leads

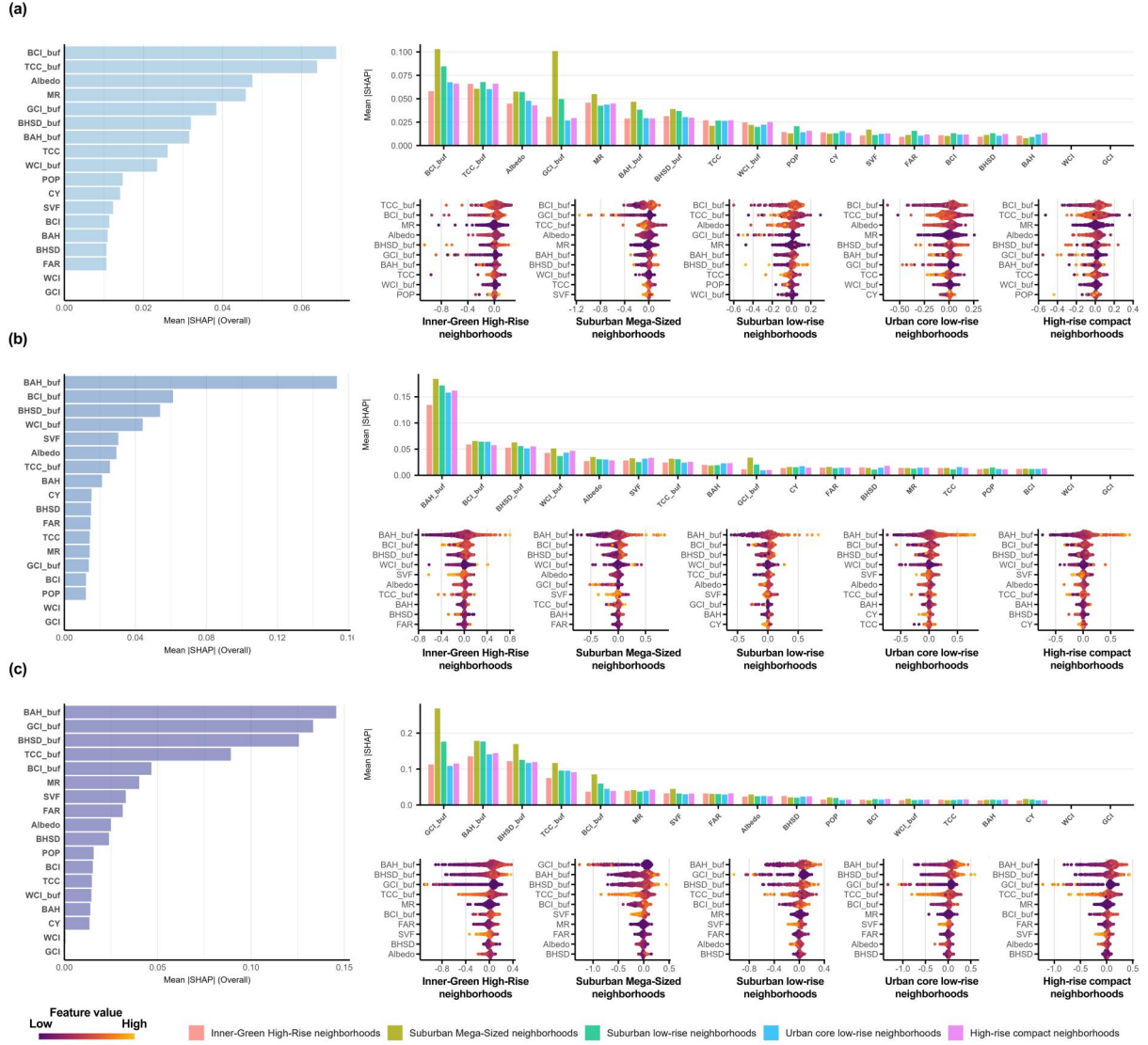


Figure 8: Relative importance and summary plots of different variables during nighttime periods. (a) 19:59; (b) 00:16; (c) 04:04.

to a continuous decrease in the SHAP value. However, at 00:16, an increase in BAH_buf leads to an increase in the SHAP value. Similarly, BHSD_buf also shows a strong cooling effect at 15:40, but during transitional periods and nighttime (07:08, 00:16, 04:04), BHSD_buf more often exhibits a warming effect.

The internal environment of neighborhoods also has a significant impact on LST. Albedo is an important cooling factor during the daytime (07:08–15:40) and transitional periods (19:59), showing a strong negative correlation with the SHAP value. Internal tree canopy coverage (TCC) and building coverage (BCI) are also key factors. For example, at 13:49, the SHAP value decreases with increasing TCC, while the effect of BCI increase on the SHAP value shows some nonlinear relationships.

3.4. Heterogeneity effect of key factors on daytime and nighttime LST

We use interaction analysis to explore the heterogeneity effect of key factors. The effect sizes to indicate the direction (positive for warming, negative for cooling) and magnitude of the influence of each factor. First, we estimate the overall effect sizes of these variables on LST. The overall results (Table S3 and Table S4) show the significance and effect of the variables in the models. We find that TCC shows no significance at 00:16. Second, as shown in Fig.10, we used forest plots and post-hoc multiple comparisons to compare the effect sizes of different key variables in various subgroups with respect to the overall effect sizes.

For TCC, during the day, Compared to the overall effect size (Grey line), TCC shows a lower effect size in Suburban Mega-Sized neighborhoods. Moreover, in Urban core low-rise neighborhoods and High-rise compact

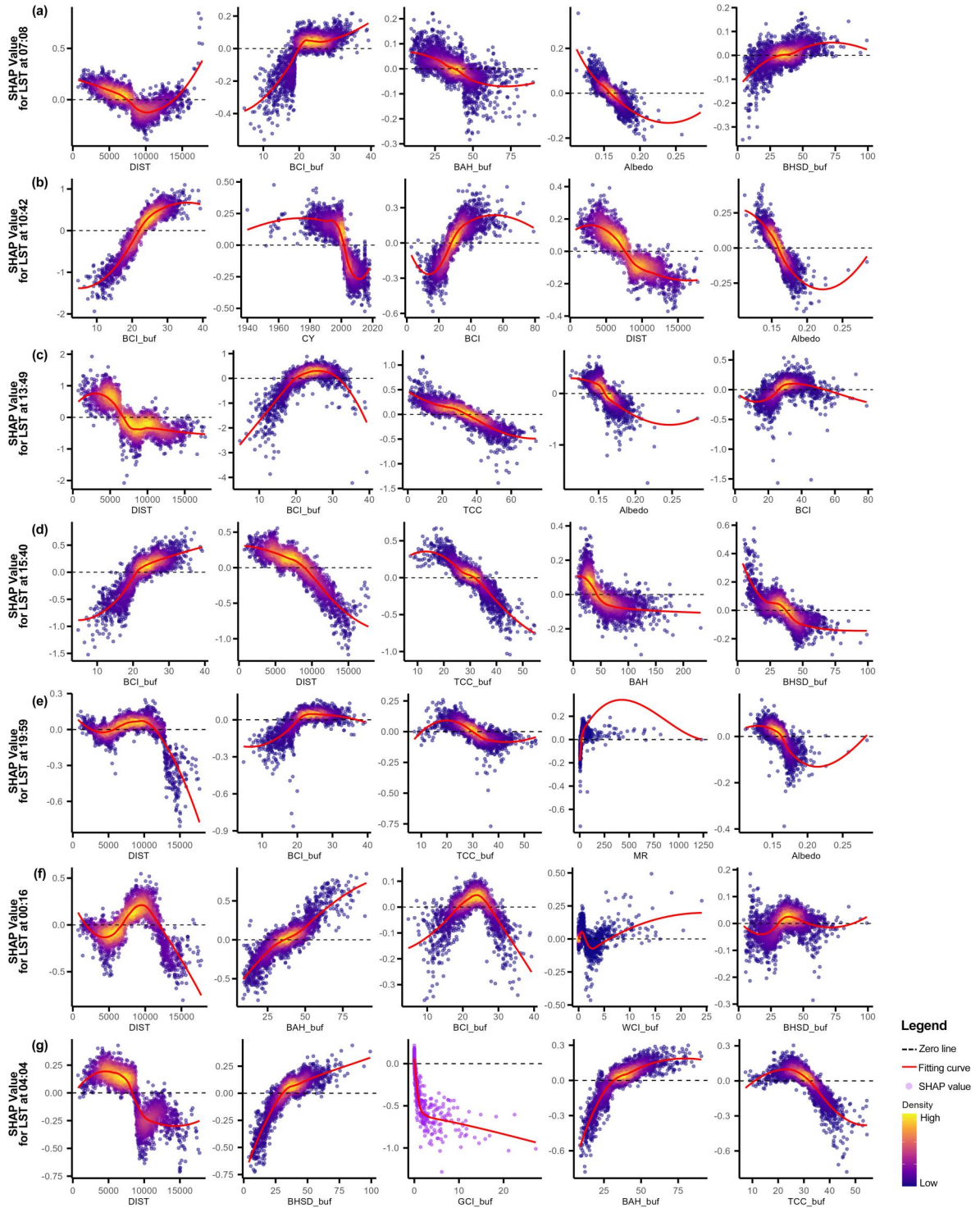


Figure 9: SHAP dependence plots of key internal and external factors across seven time periods. (a) 07:08; (b) 10:42; (c) 13:49; (d) 15:40; (e) 19:59; (f) 00:16; (g) 04:04.

neighborhoods, TCC is not significant in some time periods, and the effect size of TCC is higher compared to other neighborhoods. For example, in the High-rise compact neighborhoods, TCC is not significant at 07:08, 13:49, and 15:40 and shows a slight warming effect at 10:42. At night, it is not significant in most groups. For TCC_buf, it exhibits a significant cooling effect in most groups during the daytime. Compared to the overall effect size, TCC_buf shows a lower effect size in Inner-Green High-Rise neighborhoods at 10:42 and 15:40. During nighttime, the effect size of TCC_buf is lower in High-rise compact neighborhoods and Urban core low-rise neighborhoods at 19:59.

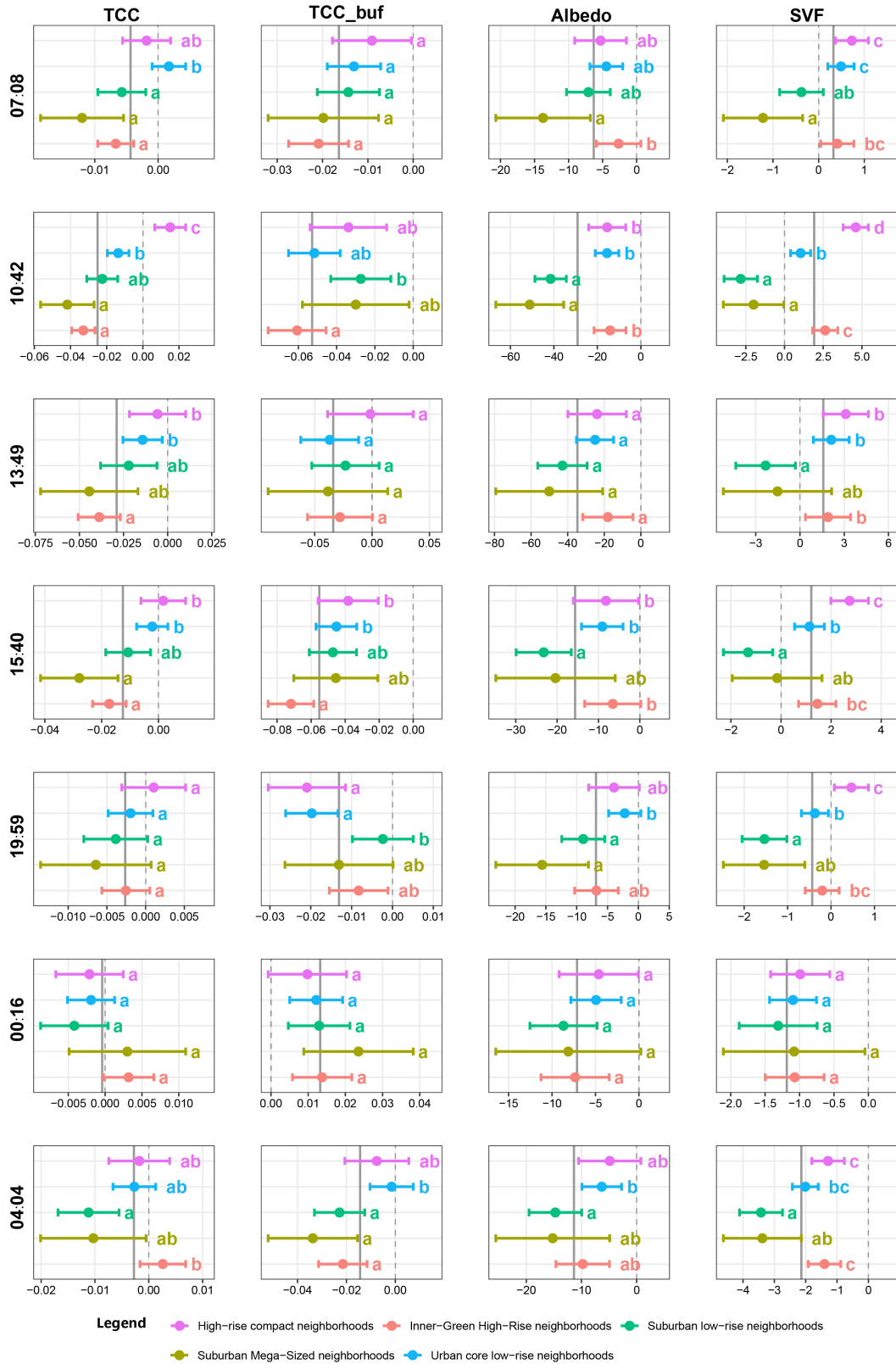


Figure 10: Differences in the effects of TCC, TCC_buf, Albedo, and SVF across different types of neighborhoods under the seven temporal scenarios. Groups without shared the same letter are considered to have significant differences. Grey lines represent the average effect. Grey dash line represents the null line. If the confidence interval of a variable crosses the null line, the variable is not statistically significant in that group.

For Albedo, it shows significant cooling effects in most types of neighborhoods across most time periods. Compared to the overall effect, Suburban Mega-Sized neighborhoods and Suburban low-rise neighborhoods achieve better

cooling effects at 10:42, 15:40 and 19:59, while the cooling effect is significantly weaker in High-rise compact neighborhoods and Urban core low-rise neighborhoods at 10:42. For SVF, it shows opposite effects in different types of neighborhoods. During the daytime, SVF shows cooling effects in Suburban Mega-Sized neighborhoods and Suburban low-rise neighborhoods, while it mainly shows warming effects in High-rise compact neighborhoods, Urban core low-rise neighborhoods, and Inner-Green High-Rise neighborhoods, with the warming effect being more prominent in High-rise compact neighborhoods at 10:42 and 15:40. During nighttime, SVF shows cooling effects in almost all types of neighborhoods, with stronger cooling effects in Suburban Mega-Sized neighborhoods and Suburban low-rise neighborhoods at 19:59 and 04:04.

4. Discussion

4.1. Differences among various types of neighborhoods in LST

This study proposes a refined classification method for neighborhood thermal environments. It provides a more comprehensive perspective on the differences between various neighborhoods. The study found that at night, although the differences are statistically significant, the gaps in LST are generally close. During the daytime from 10:42 to 15:40, the LST among different neighborhood types shows significant and large differences. The median LST across different neighborhoods generally follows: Urban core low-rise neighborhoods > High-rise compact neighborhoods > Inner-Green High-Rise neighborhoods > Suburban Mega-Sized neighborhoods $\approx$ Suburban low-rise neighborhoods.

This is consistent with the findings of other research. Liu et al. [42] found that most of Beijing's heat island areas are concentrated in the urban core area. Liu et al. [14] found that most compact high-rise and compact mid-rise neighborhoods in summer are high-temperature areas. Zong et al. [43] revealed that tall and dense buildings may obstruct wind that could otherwise flow into the neighborhood and raise the neighborhood temperature. These reasons may correlate with the higher LST in Urban core low-rise neighborhoods and High-Rise compact neighborhoods. Suburban Mega-Sized neighborhoods and Suburban low-rise neighborhoods have lower LST, which is related to the location of the neighborhoods and Beijing's urban heat island morphology. These two types of neighborhoods are mainly distributed between the Fourth and Fifth Ring Roads, far from the urban heat island in Beijing. The intensity of the urban heat island continuously decreases toward the external suburbs [44]. In addition, neighborhoods in marginal areas have more space to arrange plants both externally and internally, which also helps reduce LST. Our findings reveal the heterogeneity of thermal environments across neighborhoods, highlighting the need to identify neighborhood types that endure worse summer heat and to develop targeted mitigation strategies.

4.2. Impact of internal and external factors on LST in daytime and nighttime

The straight-line distance to the city center is one of the most important factors; neighborhoods with a shorter distance to the city center exhibit higher predicted LST. Our results are consistent with current research. Zhao et al. [6] have found that the distance to city center is predominantly negatively correlated with the LST. Zhou and Liu [45] have found that urban heat islands have a clear gradient effect, a phenomenon particularly prominent in highly developed cities. Usman et al. [46] have also confirmed that the nighttime LST in urban centers during summer is much higher than in rural areas. This result illustrates the importance of location. When assessing the thermal environment of a neighborhood, the urban-rural gradient is a key factor that must not be overlooked.

Revealing the impact of the external environment on the internal thermal environment of neighborhoods is also one of the major contributions of this study. BCI_buf and TCC_buf show a similar effect on the internal environment during the daytime. The impact of trees and buildings is consistent with existing research [14, 47, 48]. BAH_buf and BHSD_buf show distinct patterns between day and night. During the daytime, high-rise buildings create deeper shadows that reduce direct sunlight on the ground [49] and enhance ventilation and heat dissipation via the canyon effect. At night, however, they function as thermal storage, storing heat and releasing it more slowly [19]. Notably, external environmental factors show great relative importance during both the daytime and nighttime. This finding is consistent with existing research. Chen et al. [10] found that the thermal effect of adjacent three-dimensional urban landscapes is three times that of internal landscapes. Lin et al. [19] found that the

external environment of local climate zones(LCZs) plays a significant role in the LST within LCZs, and the impact of external factors is always stronger than local factors.

Moreover, albedo is an important predictor of lower LST. Onel et al. [50] also found that reflectivity is often negatively correlated with LST in urban built-up areas. Ko et al. [21] found that the field deployment of high-albedo pavement can reduce daytime LST by up to 5°C, and LST in the early evening (20:00–22:59) is still significantly reduced by 0.19–0.20°C.

In summary, DIST is one of the main contributors to LST in neighborhoods. In the daytime, both internal and external environmental factors are key contributors to LST. In the nighttime, however, the three-dimensional building morphology and land cover surrounding the neighborhood become the dominant contributors. Furthermore, the external building morphology exhibits opposite contribution directions between daytime and nighttime. Understanding these mechanisms helps inform targeted strategies to improve the thermal environment of neighborhoods.

4.3. *Heterogeneous effects of factors in various types of neighborhoods*

Interaction analysis reveals heterogeneous effects of four factors: TCC, TCC_buf, Albedo, and SVF across neighborhood types. In Suburban Mega-Sized and Suburban low-rise neighborhoods, albedo exhibits stronger cooling effects, while SVF negatively correlates with LST. In Inner-Green High-Rise neighborhoods, TCC_buf shows a stronger cooling effect. In Urban Core low-rise and High-rise compact neighborhoods, TCC_buf and albedo have a significant negative effect on LST, whereas internal TCC shows a weaker cooling effect or an insignificant correlation. SVF has a positive effect on LST.

TCC exhibits limited cooling effects in the two types of neighborhoods with high building coverage, which is consistent with existing research findings. Liu et al. [51] found that when the proportion of artificial surfaces(AS) exceeds a certain threshold, the warming effect of AS will outweigh the cooling effect of vegetation. Luo et al. [52] found that the increased urban surface roughness caused by compact low-rise buildings limits the natural ventilation potential of buildings and hinders the cooling effect of tree canopies. This study finds that TCC_buffer shows a better cooling effect in Inner-Green High-Rise neighborhoods. Previous research also suggested that external tree canopies cool the ambient air, and the stronger heat convection transfer capacity generated by the tall buildings in the neighborhood can introduce cool air into the neighborhood [53]. This may also be because TCC_buf modifies the microclimate conditions around the neighborhood, thereby indirectly reducing the temperature within the neighborhood.

Surface albedo achieves better cooling effects in Suburban Mega-Sized neighborhoods and Suburban low-rise neighborhoods. This conclusion is also consistent with previous findings. Liu et al. [54] also found that the strategy of increasing albedo achieved better cooling effects in LCZs in the urban marginal area. However, in dense central urban areas, high-albedo surfaces reflect more radiation onto external buildings, causing temperature increases, which may partially offset the cooling effect of high-albedo surfaces [55]. SVF shows opposite directions of influence among different types of neighborhoods during the daytime. This may be because Suburban Mega-Sized neighborhoods and Suburban low-rise neighborhoods are located in cooler suburbs. Areas with high SVF receive less ground radiation from relatively cooler skies than from hotter buildings [56]. In contrast, Urban core low-rise neighborhoods and High-rise compact neighborhoods are located in hotter urban central areas. An increase in SVF will cause building surfaces within it to absorb more heat, leading to warming [57].

In summary, due to differing background environmental conditions, these key factors play distinct roles across neighborhood types. Therefore, it is necessary to compare and identify the most effective cooling measures tailored to each neighborhood type and implement them efficiently to improve the neighborhood thermal environment.

4.4. *Practical suggestions to improve thermal environment in neighborhoods*

With the implementation of urban renewal projects in China [58], the renewal and construction of neighborhood environments have become more pressing issues. Therefore, we should examine and understand the thermal environment of urban neighborhoods from a more comprehensive perspective and provide practical suggestions to guide neighborhood planning and design, enhancing residents' well-being.

Among the five types of neighborhoods, Urban core low-rise neighborhoods and High-rise compact neighborhoods account for 36.5% and 16.8% in this study and face the hottest thermal environment during the day, posing

a threat to the large number of residents living there. However, renovating the thermal environment of these neighborhoods poses a challenge. These neighborhoods are strongly affected by the urban heat island effect or lack large open spaces for planting trees. A finding in this study also indicates that simply increasing internal tree canopy coverage (TCC) is ineffective. Instead, our results reveal that increasing external tree canopy coverage (TCC_buf) and surface albedo still has a significant cooling effect on LST in neighborhoods. Designers can also use street green spaces to increase the planting density of street trees around neighborhoods. Constructing street parks and building new parks around neighborhoods are both effective methods for indirectly reducing neighborhood LST. Designers can also renovate roads and squares using high-albedo paving materials, use porous grass pavement for parking lots, and apply cool roof technologies, including high-albedo roof coatings and green roofs, to reduce neighborhood LST. Additionally, increasing internal tree coverage and appropriately reducing sky openness is also recommended by renovating sun-exposed spaces with trees or sunshades.

Inner-Green High-Rise neighborhoods (26.1%) have tall buildings and high green coverage but face moderate thermal stress. The most effective cooling strategy is to increase external tree canopy coverage, which outperforms simply enhancing internal TCC. Therefore, we advise designers to increase external tree coverage first.

Suburban Mega-Sized and Suburban low-rise neighborhoods are located farther from the urban core and face lower thermal risks. Among them, Suburban Mega-Sized neighborhoods offer a model for future development: building large, dense neighborhoods away from the city center reduces residents' heat exposure. Increasing internal tree canopy coverage in Suburban Mega-Sized neighborhoods and increasing surface albedo in the two types of neighborhoods will be more effective methods compared to other types. Designers can consider preserving open spaces and increasing SVF to promote ventilation both inside and outside the neighborhood, thereby cooling it.

4.5. Limitations and future research

This study provides a refined classification method for urban neighborhoods that integrates internal and external built environments, location, and socioeconomic status. It can distinguish thermal environments across neighborhoods and offer targeted renovation suggestions. However, several limitations exist.

First, LST serves only as a proxy for the thermal environment. Although LST has been widely used in urban thermal environment research due to its easy availability and high resolution, it does not capture actual human thermal stress. Furthermore, due to the limitations of ECOSTRESS data, the LST acquired in this study does not cover a complete diurnal cycle. Although we selected LST images under clear-sky conditions as much as possible, temporal and other meteorological factors may still influence LST to some extent. We recommend using microclimate simulation models or long-term in-situ measurements to obtain continuous time-series climate data.

Second, although this study comprehensively considered internal and external influencing factors, some potentially important influencing factors may not have been included. The regression equations constructed in this study are not optimal. Model performance (R^2) was suboptimal during certain periods. We advise future work to consider 3D vegetation characteristics and green space patterns and to test alternative models.

Third, Although this study shows a nonlinear relationship among variables. The subgroup comparison method used in this study only compares effect sizes and cannot estimate or compare the complex nonlinear relationships that may exist for these variables in different types of neighborhoods. Future studies could try comparing the non-linear relationships between different groups.

Finally, the study area is limited to the Fifth Ring Road area of Beijing, and only LST images from seven time periods in summer were studied. Subsequent research could extend the study to the urban thermal environment in different seasons and conduct comparative studies across multiple cities to further verify, extend, and generalize the conclusions drawn in this study.

5. Conclusion

This study proposes a classification method that comprehensively considers neighborhoods' internal and external built environments, reveals fine-grained categories of neighborhoods within the urban core area, and explores the heterogeneity of thermal environments among different types of neighborhoods, providing targeted cooling measures for each type of neighborhood. The main findings are as follows:

(1) The study area was divided into five types of neighborhoods. The gap in LST among neighborhoods in the nighttime is close, but there are significant differences in daytime (10:42-15:40) LST among different types. Overall: Urban core low-rise neighborhoods > High-rise compact neighborhoods > Inner-Green High-Rise neighborhoods > Mega-Sized neighborhoods $\approx$ Suburban low-rise neighborhoods.

(2) DIST was identified as the most influential factor for daytime and nighttime LST in neighborhoods. During the day, both internal and external landscape factors make significant contributions to neighborhood LST; at night, external landscape factors generally contribute more than internal factors. The key factors with high importance have nonlinear relationships with LST during seven time periods.

(3) In different types of neighborhoods, the effects of internal and external tree canopy coverage (TCC, TCC_buf), surface albedo (Albedo), and sky openness (SVF) differ. TCC exerts better cooling effects in Suburban Mega-Sized neighborhoods during the daytime. TCC_buf can exert a cooling effect in all types of neighborhoods, with a better cooling effect in Inner-Green High-Rise neighborhoods. In Suburban low-rise neighborhoods and Suburban Mega-sized neighborhoods, Albedo exerts a better cooling effect, and SVF is negatively correlated with daytime LST.

In summary, the results of this study reveal the complexity of neighborhood environments, provide an executable scheme for fine-grained neighborhood classification, comprehensively explore the influence of internal, external, and location factors on the neighborhood thermal environment, and offer deeper insights and practical suggestions for mitigating the urban heat island effect and neighborhood renewal efforts.

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Data availability

Data of this study is available at <https://github.com/juanfenger/dataspace-manuscripts-3>.

Declaration of Competing Interest

The authors declare no conflict of interest.

Author contributions

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